

Business Technology Management 415

Project Report

Vince Davis

BTM 415 - B01

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Table of Contents

1. Motivation.....	2
2. Database Design and Construction.....	6
3. Data Visualization.....	11
4. Summary.....	19

1. Motivation

Our team began by evaluating potential project topics across finance, technology, and sports. The decision ultimately centered on sports, specifically basketball, driven by both strong team interest and significant data availability. A primary motivator was the genuine passion one of our members holds for basketball. This enthusiasm was complemented by the discovery of extensive open-source data related to the sport. We found this data to be not only abundant but also remarkably clean, well-organized, and spanning a significant historical period from 1946 to 2023. This accessibility presented a clear advantage compared to the anticipated difficulty in acquiring similarly detailed data for other topics, such as technology sector metrics like phone sales. Beyond the practicalities of data acquisition, basketball emerged as an important and engaging topic. Its relevance is underscored by its large cultural and economic footprint globally. Moreover, it serves as a prominent and dynamic field for the application and advancement of data analysis methodologies, making it a valuable case study. Interestingly, initial assumptions of simplicity were quickly dispelled as we uncovered the considerable depth involved, with numerous complex factors affecting game scores and player performance. This complexity is mirrored in the data's variety which goes far beyond basic scoring statistics, offering diverse information types that make the analytical process inherently interesting.

Our team chose the NBA as the focus of this project because it offers a rich and dynamic dataset that reflects evolving trends, strategic decisions, and player development over time. The league continuously generates interest not only among fans but also among analysts, coaches, scouts, and decision-makers who rely heavily on data-driven insights.

The central motivation behind this project is to explore how data visualization and database design can provide a clearer, more comprehensive understanding of the NBA landscape.

Specifically, our project aims to answer the following key questions:

1. How has the shot selection changed from the 2021-22 to the 2022-23 season?
2. Which positions (PG, SG, SF, PF, C) contribute the most to different statistics (Points, Assists, Rebounds, Steals, Blocks)?
3. What is the relationship between free throw attempts and winning percentage?
4. What are the top colleges that produce today's NBA players?
5. Which NBA players have the highest salaries?
6. How does height affect rebounding ability?
7. Which teams have the biggest home-court advantage (which home arenas produce the highest win percentage)?

In evaluating our project's contribution relative to existing information, we recognize that some of our research questions primarily involve reporting widely available public data. For example, identifying the highest paid NBA players or the top colleges producing current players compiles facts readily accessible through major sports media and databases. While useful for context within our report, these specific findings reiterate existing knowledge rather than providing novel information. However, the core analytical components of our project aim to offer additional value beyond commonly cited statistics or general discussion. We focus on providing detailed, quantitative insights specifically for recent NBA seasons. For instance, our examination of the change in shot selection between the 2020-21 and 2021-22 seasons seeks to present a similar breakdown, potentially offering more detail than typical high level media

summaries. Similarly, while the general statistical roles of different player positions are well known, our work quantifies these contributions precisely using current data, which may highlight subtle shifts or nuances. Furthermore, when exploring established relationships, such as between free throw attempts and winning percentage or height and rebounding ability, our contribution lies in empirically quantifying these connections using current data, offering an updated perspective rather than relying solely on historical understanding or anecdotal evidence. Our home court advantage also focuses on providing an updated quantitative ranking based on recent performance data. Therefore, while not all aspects of our research unearth entirely new phenomena, the project seeks to add value by applying quantitative methods to specific, relevant questions about the current state of the NBA. The goal is to provide detailed data driven insight into recent trends and performance metrics, moving beyond surface level observations.

The findings generated by this project hold relevance for several distinct groups interested in NBA analytics and performance trends. By providing detailed, quantitative insights derived from recent league data, our analysis can inform and benefit various stakeholders seeking to understand the nuances of the modern game. Professionally, NBA teams, encompassing front offices, coaching staffs, and internal analytics departments, represent a key stakeholder group. Insights drawn from our analyses, such as the specific shot selection shifts between 2020-21 and 2021-22 seasons, current positional contributions to key statistics (Points, Rebounds, etc.), the quantified relationship between factors like free throw attempts and winning percentage, and updated home court advantage rankings could serve as valuable supplementary information. While teams possess their own extensive data, external analyses can offer broader league context, validate internal findings, or highlight trends relevant to strategy and player evaluation. Equally, sports media entities and analysts, which includes journalists, broadcasters,

and content creators will be benefited. Our findings provide concrete data points and quantitative backing for their narratives, enriching articles, commentary, and analytical pieces with updated information on player roles, league trends, and team performance factors like home court strength. Beyond the professional sphere, engaged fans interested in a deeper understanding than typical box scores provide can benefit from exploring the tactical shifts, the statistical impact of different positions , and factors like height's influence on rebound or the variability in home court success. Such insights can enhance their appreciation of the game's complexities. The quantitative nature of our findings on players' contributions and factors linked to team success may also be of particular interest to participants in the fantasy basketball league and sports betting markets. Furthermore, academic researchers and fellow students focusing on sports analytics or data science could find value in our methodology and results based on recent data, and the project as a practical case study. Ultimately, these diverse groups could benefit from the project's focus on delivering specific, data driven insights into the dynamic of the NBA, moving beyond anecdotal evidence or generalized observations.

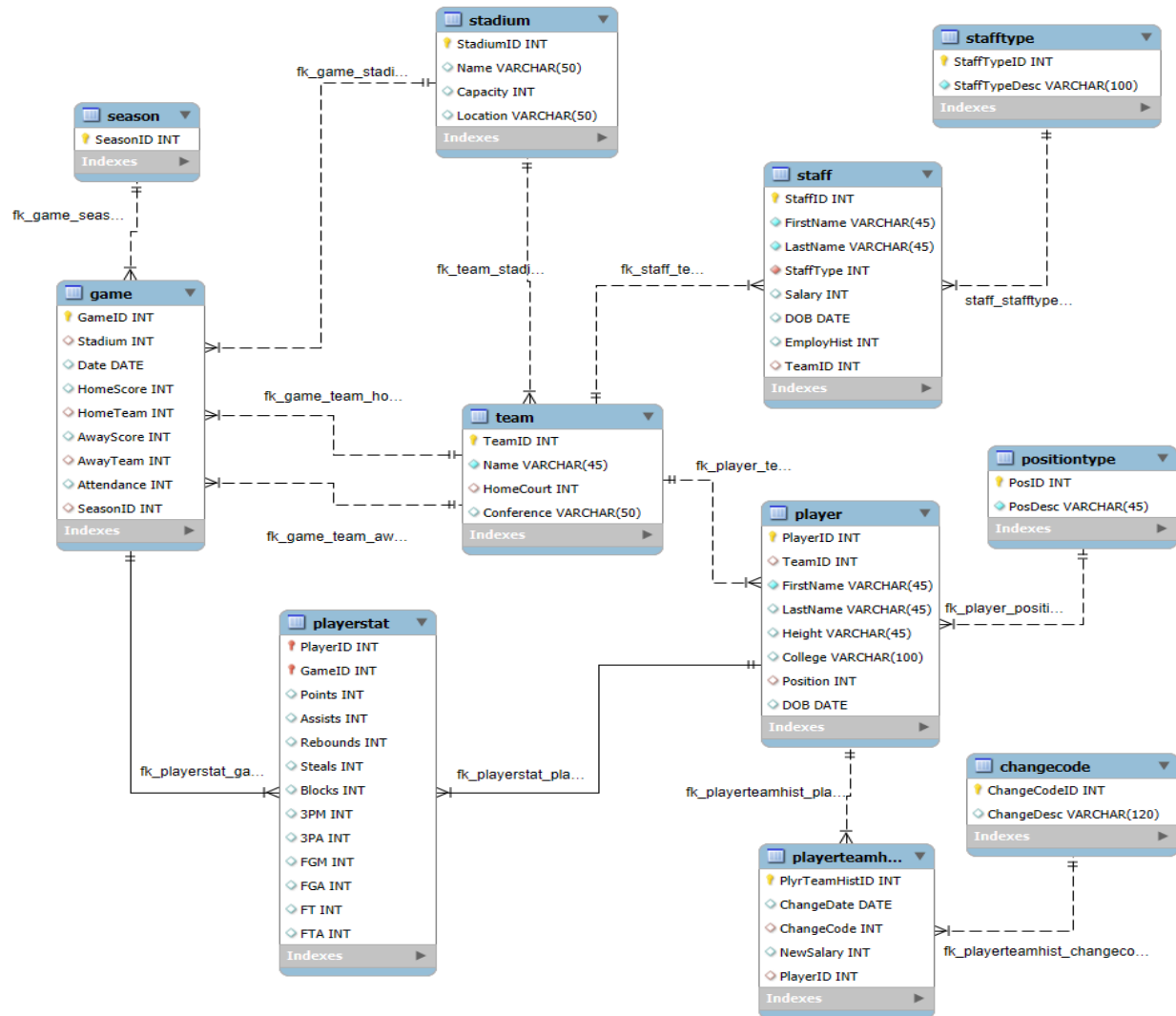
2. Database Design and Construction

For this project, we collected data from three primary sources: Basketball Reference, Kaggle, and Sportradar. **Basketball Reference** (<https://www.basketball-reference.com/>) is a widely respected and comprehensive database for professional basketball statistics. The platform offers reliable and well-organized data, which we exported directly into spreadsheet format for ease of use. **Kaggle** (<https://www.kaggle.com/datasets/wyattowalsh/basketball/data>) provided us with structured NBA datasets focused on players, teams, and box scores, many of which are already formatted for use with SQL, ensuring data consistency and quality. **Sportradar** (<https://developer.sportradar.com/basketball/reference/nba-overview>), as the official source of NBA statistics, was also referenced to validate schema design and API structure. After gathering the data, we used Microsoft Excel to perform preprocessing, which included removing irrelevant columns, standardizing headers, and filtering the dataset to include only games played between October 2020 and April 2023. We specifically excluded games without recorded attendance due to COVID-19 restrictions, as these would skew the analysis of home-court advantage. Our database is structured around 11 relational tables, although not all were imported into a MySQL database:

- **changecode**: Records when players change teams during or between seasons, including change code ID and descriptions.
- **game**: contains details for each NBA game, including stadium, date, home/away teams, home/away scores, and attendance.
- **player**: stores player demographic information such as player ID, team ID, name, height, position, and birthdate.

- playerstat: captures individual player performance stats per game (points, rebounds,...etc)
- playerteamhist: tracks each player's history with different teams and seasons, useful for understanding player movement over time.
- positiontype: defines standard NBA position roles with position ID.
- season: contains season identifiers.
- stadium: include data on NBA stadiums such as name, capacity, and location.
- team: stores team-level information including team ID, name, home court, and conference.
- staff: intended to store information on non-player personnel such as coaches and trainers, but remains empty due to lack of available data.
- stafftype: intended to classify staff roles, but no data was imported for this project.

To better illustrate the structure and relationships within our database, we generated a reverse-engineered logical model based on the final schema in MySQL. This diagram visually represents how the tables are connected and highlights the primary keys, foreign keys, and key relationships between entities:



This diagram represents a reverse engineered logical model of the relational database schema that was designed for our NBA analysis project. It provides a detailed view of the database's structure and organization. The diagram illustrates the names of each table, such as season, stadium, team, player, and playerstat, along with the attributes or columns within those tables, including their data type Name as VARCHAR(45). Furthermore, the diagram emphasizes the relationships between the tables, visually represented by connecting lines. It also highlights the primary keys, which are crucial for uniquely identifying records within a table, and foreign keys,

which establish the links and connections between the tables, and foreign keys, which establish the links and connections between the tables, showing how different entities like players, teams, and games are related within the database.

To access the MySQL database and run queries for analysis and visualization, the following server and login credentials were used:

- Static IP address: 34.57.41.216
- Port: 3306
- Username: testuser2
- Password: (aaA\$sevensix\$123)

Our project necessitated developing capabilities with database management, specifically regarding data ingestion, that went significantly beyond the methods covered in lectures. While the lectures provided a foundation in database design and basic data import, using tools like SQL Workbench with smaller datasets, we encountered substantial limitations when working with the large-scale, real-world data required for our basketball analysis.

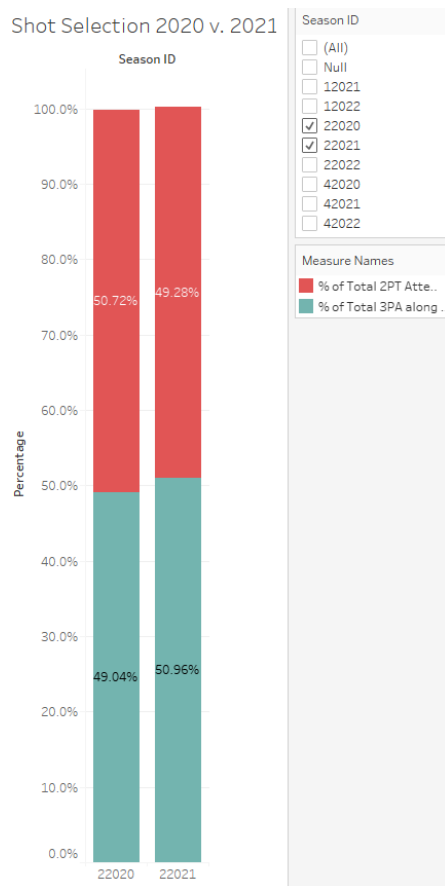
Initially, we attempted to import our data (sourced in CSV and SQLite formats) using the standard SQL Workbench table data import wizard. This approach failed when dealing with larger datasets, files around 300MB (~30000 rows) caused the application to freeze or become unresponsive. Our full datasets included SQLite files significantly larger, such as one approximately 2.5GB file representing over a million rows of data, making the import wizard approach unfeasible. Faced with this obstacle, we explored alternative solutions. After considering and discarding the idea of writing a custom C# program for row-by-row insertion due to the anticipated time investment, we developed a practical multi-stage workflow. Firstly,

we learned to utilize the `sqlite3` command line tool to handle the initial data extraction. By setting the mode to CSV (`.mode csv`) and using the `.output [filename.csv]` command, we successfully exported the raw data from the large SQLite tables into CSV file format, bypassing the workbench's limitations for the initial large file handling.

The next critical step involved data preparation using spreadsheet software (Excel). We opened the exported CSV files and performed essential data cleaning, where we manually removed extraneous columns of data, ensuring the structure aligned with our target SQL database schema. Crucially, after cleaning and structuring the data within Excel, we then divided and saved the processed data into multi smaller CSV files. This step was key, because it allowed us to return to the SQL Workbench import wizard originally introduced in lectures. By feeding the wizard these smaller, cleaned, and properly structured CSV files individually or in manageable batches, the tool could successfully process and import the data into our database without encountering the freezing issue caused by the original large, raw file.

The entire pipeline - diagnosing the workbench's limitations with larger files, evaluating other options like C#, using `sqlite3` command line for CSV export, performing necessary column manipulation and data cleaning in Excel on large datasets, strategically breaking the processed data into smaller manageable files, and finally adapting the standard import wizard to work with these chunks - was a process developed outside the scope of the lectures. We learned these techniques through independent troubleshooting, utilizing online resources like Google searches, technical documentation, forums, and guidance from AI assistants, combined with practical experimentation. This represented a significant, hands-on extension on our data handling skills, driven by the necessity of managing large-scale, real-world data for the project.

3. Data Visualization

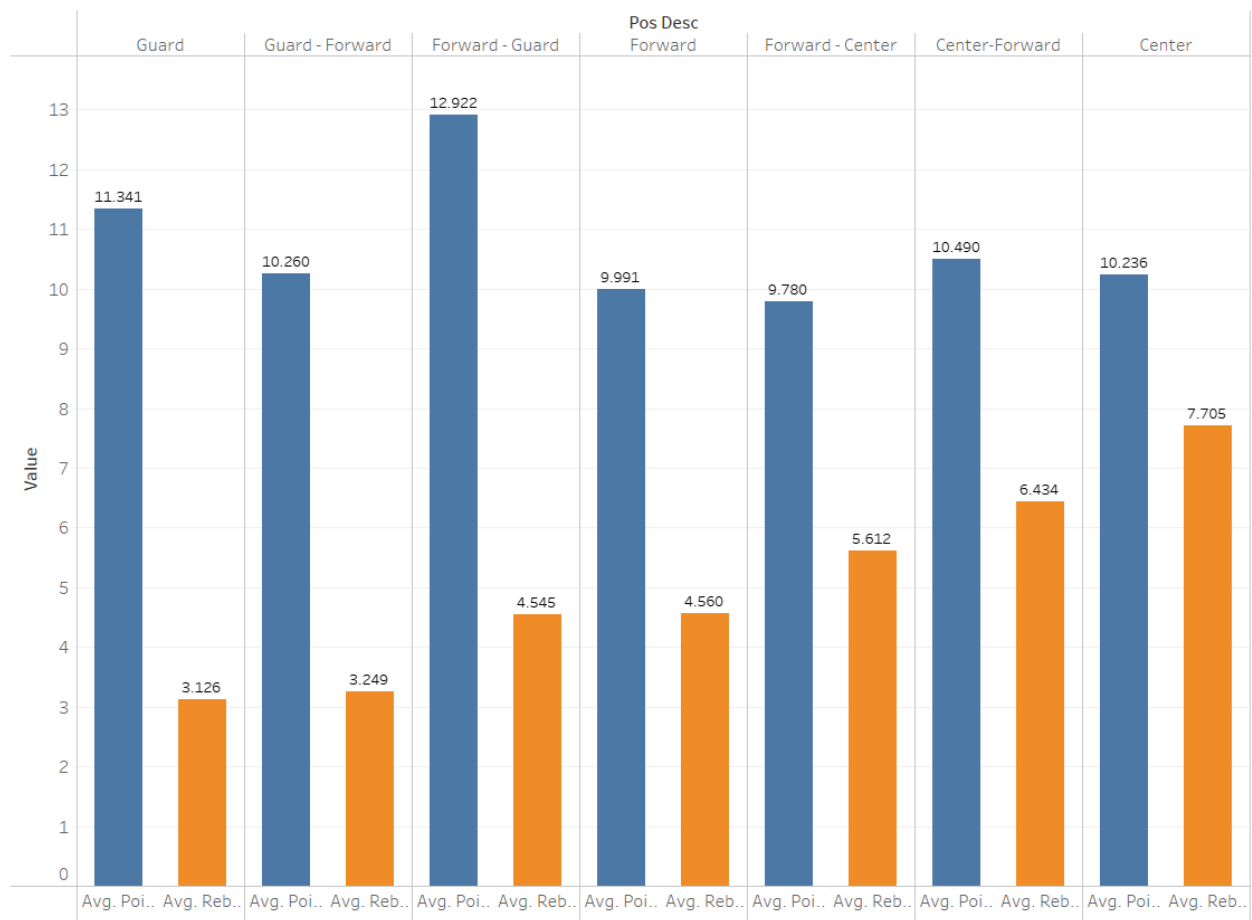


The shot selection visualization displays the percentage of 2-point shots (red) and percentage of 3-point shots (green) taken in the 2020-21 and 2021-2022 seasons. The percentage is out of a total of both 2-point and 3-point attempted shots. This visualization aimed to answer the question, “*How has the shot selection changed from the 2021-22 to the 2022-23 season?*”.

However, during the data importing process, we found difficulty in being able to import the data relevant for this question. More specifically, the data this visualization stems from is the playerstat table, which contains additional relevant attributes such as Assists, Steals, and Blocks. The data we planned to import included those attributes as well as data from later seasons, such as 2022-2023. Due to the large size of the data required to import, we were unable to

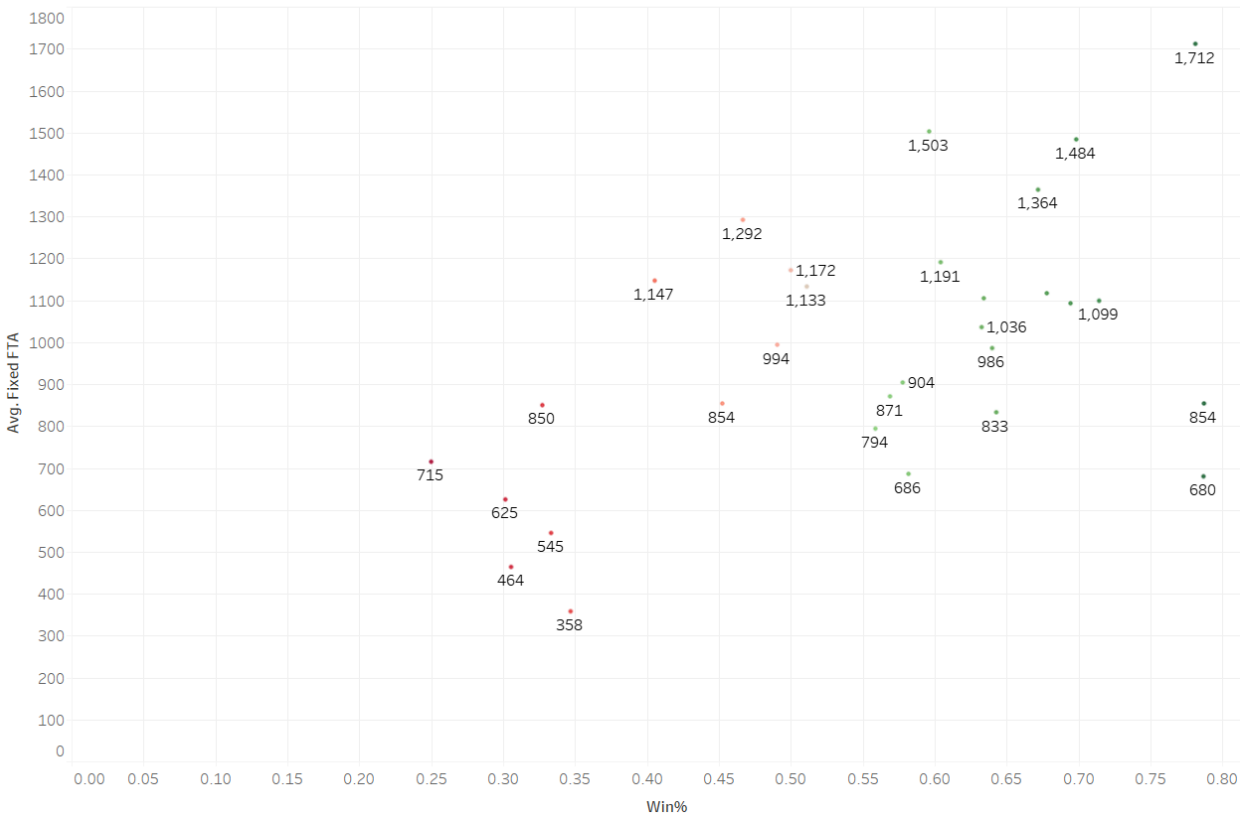
successfully import the rest of the necessary data to answer all parts of this question, even after applying the methods to work around this as described earlier. Due to this, we tailored the visualization to show the changes from the seasons that we were able to import, and decided to compare between selection of shots that were successfully uploaded. Through this visualization we found that the difference in the percentage of the 2-point and 3-point shots is minimal, meaning that the shot selection has had little to no changes between the two seasons. In the 2020-2021 season, the percentage of shots being three pointer shots was at 49.04%, and in the 2021-2022 season the percentage was at 50.96%. We found similar results in the visualization for the two pointer shots as well.

Points and Rebounds Per Position



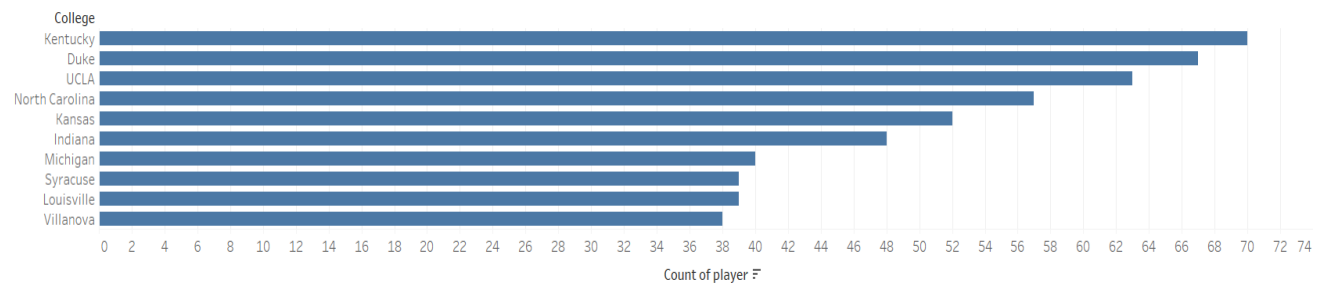
The Points and Rebounds Per Position visualization displays the disparity between the different positions and the average number of points (points per game) and rebounds (rebounds per game) within the 2020-21 - 2021-22 season. This visualization answers the question of “*Which positions contribute the most to different statistics?*”. We chose to analyze average instead of sum as the sum can skew the data. This is because there are more guards in the NBA than any other position, thus leading to a larger total, and we want to focus on the impact each position will have in the game. It is evident that forward-guards are the highest scoring position compared to the other positions. We think this is because forward-guards have the size to score in the paint, while maintaining good ball handling and good shooting abilities. Centers average the most rebounds. Rebounds linearly increase as the player positions become taller and taller (guards are shortest, centers are tallest), thus we believe height is the main factor that dictates players’ rebounding ability.

FTA Effect on Win Percentage



The FTA Effect on Win Percentage visualization depicts the correlation between free throw attempts and the win percentage of each team in the 2020-21 - 2021-2022 season. This visualization can answer questions such as “*What is the relationship between free throw attempts and winning percentage?*”. To create this visualization, a calculated field was required to calculate win percentage, and the aggregate number of free throws for each team. Win percentage is calculated by counting the number of games where the home score was greater than the away score, divided by the number of games. Free throws aggregated by team were calculated by taking the sum of free throws attempted by each team. Although there are successful teams with low free throw attempts, there seems to be a positive correlation between more free throw attempts and a higher win percentage.

Active NBA Players Per College



Count of player for each College. The view is filtered on College, which keeps 10 of 554 members.

The "Active NBA Players Per College" Horizontal Bar Chart illustrates the number of active NBA players who attended various top colleges. This visualization offers valuable insight into which institutions are most effective at developing talent for the NBA. It helps answer key questions such as, “*Which colleges produce the most NBA players today?*”. To create this chart in Tableau, we used the count function to calculate the number of players from each college. The college names were used as categories, and the data was then sorted in descending order to highlight the schools with the highest NBA representation and then we filtered “College” to get the top 10 colleges with the most players. From the visualization, it’s evident that certain colleges, most notably Kentucky and Duke, consistently produce a significant number of NBA players.

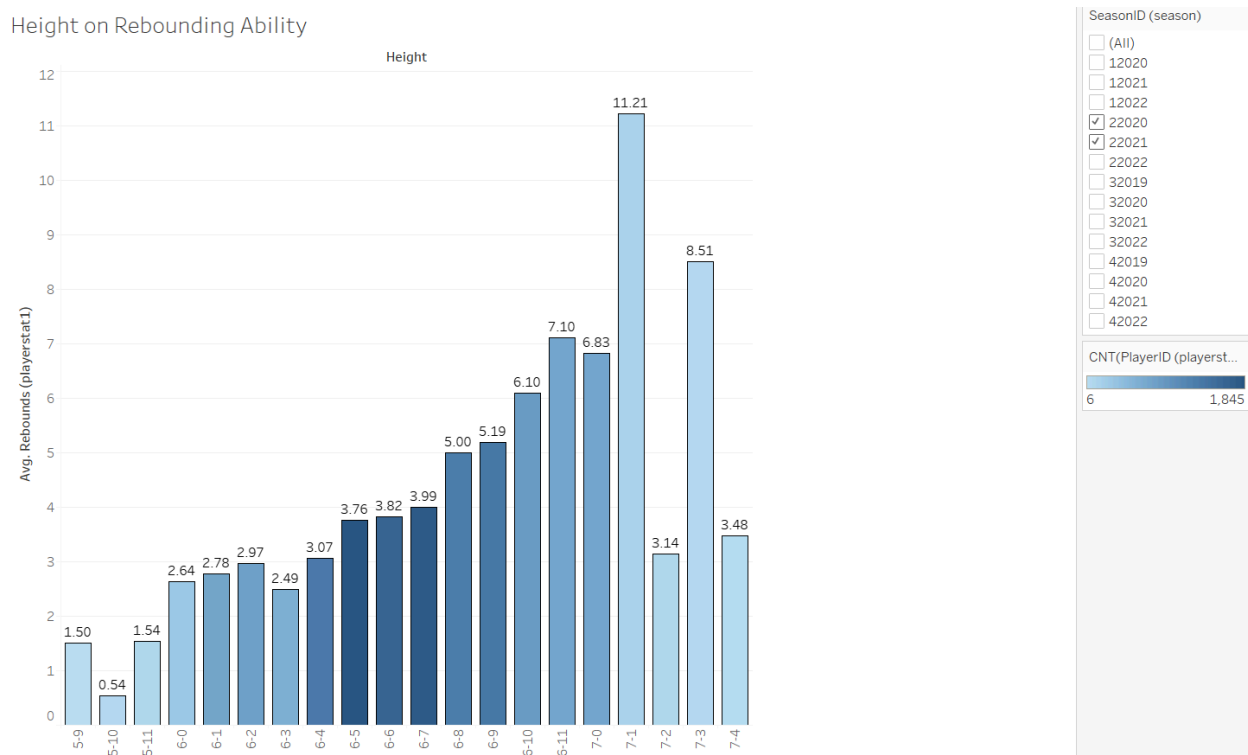
Players Salary



Sum of New Salary for each [First Name]*" "+[Last Name]. The data is filtered on New Salary, which ranges from 21382557 to 48070014.

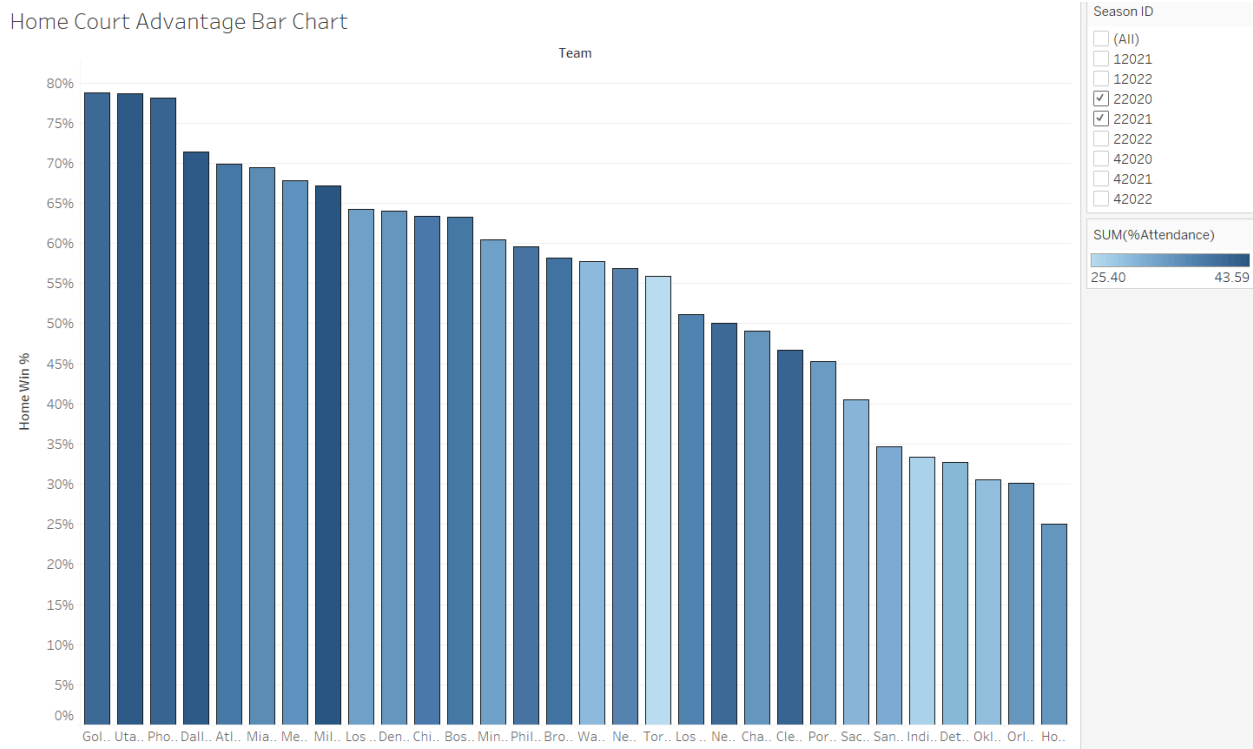
The "Players Salary" Horizontal Bar Chart visualizes the NBA players with the highest current salaries, offering valuable insight into how top talent is financially rewarded in the league. This chart helps answer the question: “*Which NBA players earn the most?*”. To build this

visualization in Tableau, we used the [First Name] + " " + [Last Name] function to combine player names into a single label. Salaries were then sorted in descending order to highlight the highest earners and then we filtered “Salary” to get top players with the highest salaries in the league. From the chart, it’s clear that Stephen Curry and LeBron James lead the rankings, with Curry earning the top spot at approximately \$48 million.



The “Height on Rebounding Ability” chart shows the average number of rebounds retrieved per game, organized by height. The visualization can answer questions like “*How does height affect rebounding ability?*”. The colour of the bars represent the number of players within that height, where the darker blue bars (such as 6-5 and 6-7) illustrate that there are more players with that height, while the lighter colour represents a fewer number of players at the specified height. Majority of NBA players are around 6-5 - 6-9. There is a gradual increase in average rebounds

per game up until 7-1. It becomes ambiguous on how many rebounds player's taller than 7-1 will obtain per game.



The Home Court Advantage Bar Chart visualizes the home win percentage for each NBA team, providing insight into how teams perform on their home courts. This visualization can be used to answer questions such as “Which teams have the biggest home court advantage?”. To create this visualization in Tableau, two calculated fields were created to determine the number of home wins and the total home games. Specifically, the “Home Win” field was calculated by assigning a value of 1 when the home score exceeded the away score, and 0 otherwise while the “Home Games” assigns a value of 1 for each home game to count total games played at home. These two fields were then used to compute the win percentage by dividing the sum of home wins by the total number of home games. From the visualization, it is clear that certain teams such as the

Golden State Warriors and Phoenix Suns had notably high home win percentages, indicating strong home-court advantage. In contrast, teams like Houston Rockets and Detroit Pistons had relatively low home win rates, suggesting they struggled to perform even when playing at home. Additionally, the chart colours also represent the percentage of the stadium that is in attendance, showing that teams with relatively low attendance lose more games (light blue colour) compared to teams with high attendance (dark blue colour). This chart effectively highlights the disparity in home performance among NBA teams.

4. Summary

This project explores how data visualization and database design can enhance the understanding of the NBA landscape. The team selected basketball as their focus due to strong interest and the availability of extensive, clean, and well-organized open source data spanning from 1946 to 2023. The project aims to answer questions related to shot selection changes, positional contributions to statistics, the relationship between free throw attempts and winning percentage, top colleges producing NBA players, player salaries, the effect of height on rebounding ability, and home court advantage.

The project uses data from Basketball Reference, Kaggle, Sportradar. The database is structured around 11 relational tables, including data on games, players, player statistics, teams, and stadiums. The Team encountered challenges in importing large datasets into MySQL and developed a multi-stage workflow to overcome these limitations.

The data visualization created for the project include: Shot Selection 2020 vs. 2021, Points and Rebounds Per Position, FTA Effect on Win Percentage, Active NBA Player Per College, Players Salary, Height on Rebounding Ability, and Home Court Advantage Bar Chart. These visualizations provide insights into various aspects of the NBA, such as player performance, team dynamics, and factors influencing success.